



# Binary protein mixture separation and purification by a cassette having an internally staged ultrafiltration membrane stack

Lixin Feng<sup>a</sup>, Yufeng Song<sup>a</sup>, Gregory Straeffer<sup>b</sup>, Andrew Bartlett<sup>b</sup>, Christina Carbrello<sup>b</sup>, Sagnik Basuray<sup>a</sup>, Kamallesh K. Sirkar<sup>a,\*</sup>

<sup>a</sup> Department of Chemical and Materials Engineering, Center for Membrane Technologies, New Jersey Institute of Technology, University Heights, Newark, NJ, 07102, USA

<sup>b</sup> MilliporeSigma, 80 Ashby Road, Bedford, MA, 01730, USA

## ABSTRACT

Recently, a mixture of hemoglobin (Hb) and bovine serum albumin (BSA) was separated into two highly purified fractions with high recovery via internally staged ultrafiltration (ISUF) having a stack of three identical flat 100 kDa ultrafiltration (UF) membranes in a stirred cell. Selectivities as high as 1000–4000+, and high recoveries of individual species were achieved. Successful separation and purification of IgG from a BSA-IgG mixture was also achieved in a stirred cell using a modified ISUF technique with two 100 kDa membranes on a 70 kDa membrane; here BSA was a model for host cell proteins (HCPs) in post-protein A eluate. We explored here separation and purification of the Hb-BSA system in a developmental cassette of 88 cm<sup>2</sup> membrane surface area, having a stack of three 100 kDa membranes on each side of the channel. The channel gap was smaller than that in regular cassettes resulting in higher flow pressure drops. The Hb-BSA separation was studied over a diavolume (DV) range of ~0–6. The Hb-BSA selectivity was as high as 1600 and remained above 300 till 6 DV. Lower feed pressure and a particular permeate collection mode leading to reduced transmembrane pressure drop were crucial for higher performance. Species recoveries were lower for the DV range used especially compared to those in a stirred cell. Successful separation and purification of BSA-IgG system was also demonstrated with similar limitations. Increased flow pressure drop due to flow channel height reduction in existing cassette frames and lower flow rates leading to higher concentration polarization suggest modification of the cassette frame for high performance ISUF for difficult-to-separate protein mixtures in biopharmaceutical separation processes.

## 1. Introduction

Membranes and modules have been successfully developed and used in biopharmaceutical and biotechnology industries for sterile filtration, clarification, initial harvest, virus removal, protein concentration, buffer exchange, and protein purification [1]. However, ion exchange chromatography, affinity chromatography, and size exclusion chromatography are almost always employed for separation and purification of proteins or monoclonal antibodies (mAbs) from their binary/multicomponent mixtures [2]. This is due to the following limitation of ultrafiltration (UF) membranes due to their significant pore size distribution: unless the molecular weight ratio (MWR) of two proteins is at least ~7–10 [3], protein mixture separation by UF is not undertaken. Novel cascade configurations employing separate UF devices numbering two or three and separate pumps with individual devices were studied to separate protein mixtures having lower MWRs [4]. High-performance tangential flow filtration (HPTFF) technique was developed to separate protein mixtures having lower MWRs with considerably higher selectivity in one device [5–9].

To achieve high selectivity, the operating principles of HPTFF are as follows: maintain pH equal to the pI of the smaller species preferentially passing through the membrane; selected membrane's charge should repulse the larger protein with a similar charge at the selected pH; (3) maintain a low ionic strength in the buffer to avoid shielding species charges; (4) avoid gel polarization conditions and employ pressure where the flux varies linearly with applied  $\Delta P$  [5–9].

Consider now a mixture of Hb and BSA having a low MWR of 1.03. Conventional UF process will provide a very low selectivity. HPTFF technique exploiting membrane charge and species charge with a selected pH and appropriate buffer strength achieves a high selectivity: for example, in Hb-BSA separation, a selectivity of 70 was obtained [8]. However, the permeate did not have an almost pure Hb, far from it. The permeate in the internally-staged ultrafiltration (ISUF) technique [10, 11] employing a stack of three identical UF membranes without any gaskets or spacers in-between the membranes in one stirred cell contains an almost pure Hb [11]. In the ISUF technique, the permeate from the first membrane highly enriched in Hb via HPTFF is imposed as the feed to the second membrane whose permeate becomes even more purified in

\* Corresponding author.

E-mail address: [sirkar@njit.edu](mailto:sirkar@njit.edu) (K.K. Sirkar).

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Hb and is imposed as a feed on the third membrane to yield an even purer permeate in Hb. A number of subsequent studies [12–16] have investigated a variety of protein separations using multilayer UF membranes in one cell.

The ISUF study of Hb and BSA separation [11] achieved a high purity of Hb in the permeate even though the MWR is only 1.03; there was no study of BSA purity and recovery in the retentate, the extent of Hb recovery in the permeate and the effect of diavolume (DV) on such quantities. Since such aspects are important to determine viability of the ISUF technique, a recent study [17] illustrated achievement of high purity and high recovery of BSA in the retentate and high purity and high recovery of Hb in the permeate. This study [17] employed a stack of 3 Ultracel® 100 kDa membranes in a stirred cell achieving selectivities as high as 6000 which however depended on the DV. The selectivity is not improved by DV directly, but it changes as a function of DV. The selectivity went up with DV up to ~2–3 DVs where the maximum purity contrast was achieved, and then decreased with further DV as both fractions approached their final purified states. More importantly, DV of up to 10 was needed to recover almost all of the Hb into the permeate with very high purity; further, DV of up to 10 ensured almost complete retention of BSA with very high purity [17].

It was mentioned [11,17] that this configuration essentially changed the somewhat broad molecular weight cut-off (MWCO) profile of commercialized UF membranes to one close to a step function; particular values of pH and transmembrane pressure were essential for this result. Contrast these successful results with that of a recent one [18] of Hb-HSA (human serum albumin) separation employing four conventional hollow fiber UF modules and individual complexation of Hb with haptoglobin (Hp) and HSA with Immunoglobulin G (IgG) followed by dissociation of each complex; recovery of HSS-IgG was ~50 % and recovery of Hb-Hp was ~10–15 %.

To overcome the broad MWCO profile of current UF membranes, isoporous membranes were developed having a specific pore size via block copolymers [19]. This approach requires a particular membrane to be developed for a particular protein to be selectively transmitted as the other protein is blocked out. The ISUF approach bypasses the need for development of a specific isoporous membrane having the macrosolute-specific pore size by using a 3-membrane stack. Our original study of Hb-BSA system in Ref. [11] demonstrated clearly that a stack of 3 identical UF membranes did significantly better than a stack of 2 identical UF membranes. Therefore, in the latest publication [17], there is no information on 2-stack membranes even though we paid a passing attention to a stack of two flat membranes in a stirred cell. A significant result from these unpublished results in a stirred cell is provided in the Supplementary Information (SI) via Figure S1. It shows that a 3-membrane configuration provides a significantly higher recovery of Hb in the permeate just as it provides a considerably higher purity of BSA in the retentate.

We can however broaden the base of a 3-membrane stack. One can employ a combination of different commercially available UF membranes in the membrane stack as was done recently and successfully for IgG-BSA separation and purification [20]: the 3-membrane stack had 2 Omega® 100 kDa membranes on top of a 70 kDa Omega membrane. Earlier ISUF studies [10] in a stirred UF cell using three YM30 regenerated cellulose membranes of 30 kDa MWCO produced a purified protein in the permeate for the following systems: myoglobin and  $\beta$ -lactoglobulin; myoglobin and  $\alpha$ -lactalbumin.

There appears to be a broad base for separating and purifying various protein mixtures using ISUF. Exploring scaled-up ISUF is useful. Scaled-up UF studies recently explored single pass tangential flow filtration (SPTFF) cassettes having a long path length and often multiple channels in parallel that are being used commercially: recent examples include concentrating IgG solutions 10+ times [21], modeling such concentration processes in SPTFF cassettes [22] and inline concentration of lentivirus via model nanoparticles [23]. We study here separation and purification performance of a 3-membrane stack of Ultracel® 100 kDa

membranes in a TFF cassette with a membrane surface area of 88 cm<sup>2</sup> for the following binary mixtures: System 1, Hb and BSA; System 2, BSA and IgG (with BSA representing a model host cell protein (HCP) in post-protein A eluate).

The Hb-BSA system was selected for two reasons: It represents a system having a MWR of only 1.03 and therefore a very difficult separation. Further, it models the practical separation problem of Hb-HSA separation [18]. The BSA-IgG separation problem represents systems involving host cell protein (HCP) removal from post-protein A eluate where BSA can represent many of the smaller HCPs. BSA has also been used as a model HCP for removing process-related impurities in mAb purification via adsorptive depth filters and adsorptive hybrid filters [24] and in anion exchange membrane chromatography for effective removal of viruses needed in mAb purification with BSA mimicking acidic HCPs competitive for virus binding [25].

The cassette frame used in the present study is identical to what is used in current commercial devices. Since instead of a single membrane on each side of the channel, a stack of three membranes is being used on each side, the feed channel height is reduced somewhat leading to a significantly increased feed pressure drop for a given feed flow rate. Therefore, the transmembrane pressure drop is likely to vary significantly in the flow direction with the modified cassette of this study. The ISUF technique rides on the shoulders on HPTFF technique where the transmembrane pressure drop must ensure operation in the linear flux regime. As a result, feed flow rate, feed pressure and permeate side pressure management are factors of importance here.

## 2. Experimental

### 2.1. Materials and reagents

Hemoglobin (Hb, MW 64,677) and bovine serum albumin (BSA, MW 66,430) were obtained from Sigma (St. Louis, MO). The isoelectric points (pIs) for Hb and BSA are respectively is 6.8 [26] and 4.7 [27]. Tris hydrochloride (Tris-HCl) and Tris base were used to prepare 2.3 mM Tris buffer for pH 7.8. Bovine immunoglobulin (IgG, MW 150,000; pI, 5.5–8.3 [27–29]) and bovine serum albumin (BSA, MW 66,430; pI, 4.7 [27]) were obtained from Sigma (St. Louis, MO). Acetic acid (Sigma Aldrich) and sodium hydroxide (Sigma Aldrich) were used to prepare 2.3 mM sodium acetate buffer at pH 4.8. Then the buffer pH was adjusted by 0.1 M HCl or 0.1 M NaOH. A 0.45  $\mu$ m polyethersulfone membrane (VWR Int., Radnor, PA; now Avantor) was used to prefilter buffer and protein solutions. Sodium azide was from Sigma Aldrich.

### 2.2. Membrane cassette

The membrane cassette (Pellicon® 3 R&D RC 100kD D-Screen 02 <sup>†</sup>3 Layer) was developed by MilliporeSigma (Bedford, MA). The cassette was mounted in a regular Millipore cassette holder (Pellicon® MiniX Cassette Holder). The frontal membrane area for the device contains 88 cm<sup>2</sup> of membrane area. As in single channel conventional cassettes, it has a single feed flow channel with membrane on both sides of the channel. Unlike a conventional cassette, instead of a single membrane on each side, there are three identical 100 kDa membranes one on top of the other on each side. During preparation of the device, the structure/framework used to prepare it is identical to that used for conventional single membrane-based system; as a result, the feed channel depth was significantly reduced. The feed screen used here with Ultracel® membranes namely, “D” screen, is somewhat more open than the standard C screen so that there will be a somewhat lower pressure drop. This feed screen promotes turbulence in the feed channel enhancing the sweeping action of the cross-flow and thus improves flux in streams with fouling particles. The limitations of the feed screen are: 1) the fluid cannot contain large particles (>80  $\mu$ m), and 2) there can be a significant pressure drop in the feed channel between the feed and retentate ports due to the thickness of additional membranes reducing the channel

height. This requires further considerations during operation.

### 2.3. Experimental setup and procedure

Fig. 1 shows the cassette located in the experimental diafiltration setup and various connections. Ultrafiltration experiments were conducted using a developmental Pellicon® 3 cassette with Ultracel® Membrane in Pellicon® MiniX Cassette Holder (XX42PMINIX, MilliporeSigma, Bedford, MA). The Pellicon® 3 cassette was installed in the cassette holder according to the Pellicon® Mini and MiniX Holders User Guide on the MilliporeSigma official website. The buffer, cleaning solution, and deionized (DI) water were stored in separate reservoirs for diafiltration. All Hb-BSA runs used 1.0 mg/mL BSA, 0.2 mg/mL Hb, 2.3 mM Tris buffer at pH 7.8 (found optimal in Ref. [17]). All BSA-IgG runs employed 1.0 mg/mL IgG and 0.1 mg/mL BSA, 2.3 mM acetate buffer at pH 4.8 (found optimal in Ref. [20]).

Before an experiment, a 2.3 mM Tris-buffer solution at the specified pH was added to the tank. The cassette was thoroughly wetted and equilibrated by being flushed with 2.3 mM Tris-buffer solution at the specified pH in the diafiltration reservoir for 1 h. The pressure  $P_1$  on the feed side in the system was adjusted to 55.16 kPag (8 psig) (or another value: 34.47 kPag (5 psig)) by using a peristaltic pump on the feed side and measuring the flow rate at the retentate side. A mixed protein solution of 200 mL was added into the empty tank (Fig. 1) at the beginning up to the top which was closed. The stirring speed was set at 500 rpm in the tank. A separate tank contained 2 L buffer solution for diafiltration; it was connected to the retentate line entering the feed tank. During operation, the permeate side valve next to pressure gage  $P_4$  may be completely closed, completely open or partially open.

The volume of the permeate solution from the tank was recorded continuously; the permeated solution was pipetted into a clean cuvette. Then protein concentrations for the Hb-BSA system in the feed and permeate solutions were measured by the dual-wavelength method using a Varian Cary® 50 UV-vis spectrophotometer (Agilent, Santa Clara, CA) at 407 nm and 280 nm [10,17]. The protein concentrations in the feed solution were measured before the experiment and those in the retentate at the end of the experiment. The permeate concentrations of proteins were measured every 5 min for 30 min, then recording time was changed to every 10 min for 30 min, 30 min for 2 h, 1 h for 3 h and 2 h for 6 h. When the diavolume reached about 6, the peristaltic pump on the feed side was turned off. The concentration of the remaining protein solution was recorded via the UV-vis spectrophotometer. Protein

rejection is reported as a function of time with respect to the bulk protein concentration in the cell at that time by taking into account the total amount of protein that has permeated. Most experiments were conducted till we crossed 6 diavolume.

The protein concentrations for the Hb-BSA system and the BSA-IgG system were determined by the dual-wavelength method using a Varian Cary® 50 UV-vis spectrophotometer (Agilent, Santa Clara, CA) at respectively 407 nm and 280 nm for the Hb-BSA system [11,17] and 280 nm and 628 nm for the BSA-IgG system [20]. For testing a protein mixture at 628 nm, the protein sample was added to 3 mL of 0.1 g/L bromocresol green (BCG) solution and 1 mL of 0.5 g/L ethylenediaminetetraacetate acid (EDTA) solution for 1 h [20].

To clean the cassette after disposing of the remaining protein solution in the tank, a 0.5 % Tergazyme® cleaning solution (200 mL) was first added to the tank. The cassette was then immersed in the 0.5 % Tergazyme® solution at 50 °C for 30 min and reinstalled into the cassette holder. Using a peristaltic pump, the cleaning solution was circulated through the system for 24 h with 2 L of deionized (DI) water as if diafiltration buffer solution was being added for diafiltration except it was distilled water. After that, the cassette was soaked again in 0.5 % Tergazyme® at 50 °C for 4 h, then subjected to diafiltration with 1 L of DI water. The entire cleaning procedure mentioned above was repeated at least twice. The cassette was considered clean when the retentate flow rate and the flux of the membrane cassette reached at least 90 % of the values measured during the initial experiment with a virgin cassette. On completion of the cleaning procedure, the cassette was stored in a 0.1 % sodium azide solution at room temperature.

After cleaning, the regenerated cassette was flushed with the 2.3 mM Tris buffer at the specified pH in the reservoir for 1 h. The buffer was placed in the tank and the reservoir for diafiltration; the filtration rate of the buffer solution was recorded during this run. The regenerated cassette-based experiment was started when the filtration flow rate reached 90 % of that achieved with the previous virgin cassette-based experiment. Then protein solution filtration experiments were conducted as described earlier using the regenerated cassette. After recording the results with the rest of the protein solution, the regenerated cassette was cleaned with 0.5 % Tergazyme® solution and then uninstalled. Finally, the regenerated cassette was stored in 0.1 % sodium azide solution at room temperature. All results used the cassette regenerated using the above procedure.

For the Hb-BSA system, equations (1a) and (1b) are used to calculate respectively the rejection,  $R_i$ , and the sieving coefficient  $S_i$  of species  $i$

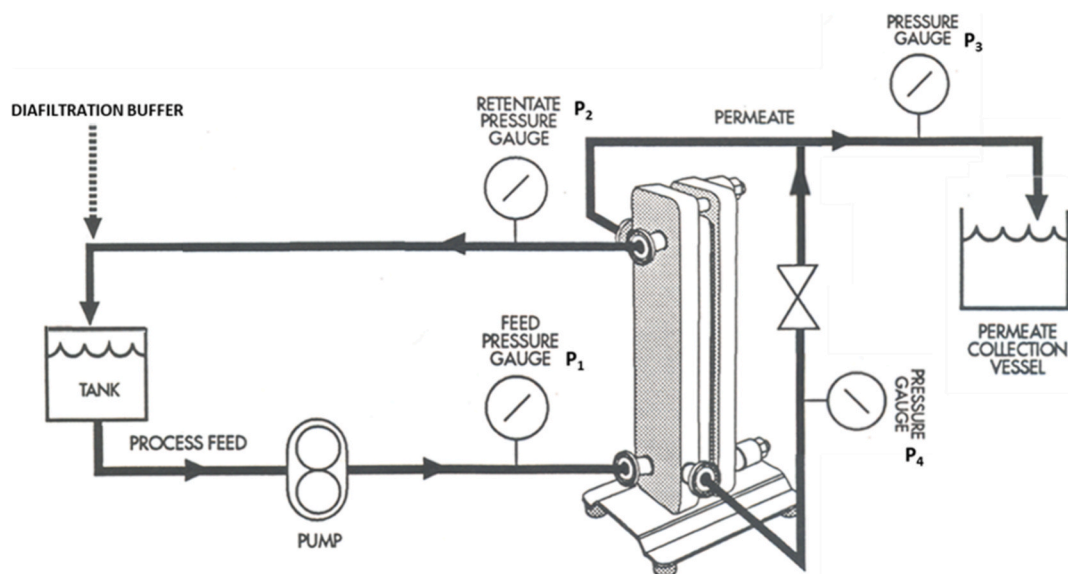


Fig. 1. The schematic of the cassette-based experimental setup for protein separations.

(Hb or BSA):

$$R_i = 1 - \frac{C_{p_i}}{C_{f_i}} \quad (1a)$$

$$S_i = \frac{C_{p_i}}{C_{f_i}} \quad (1b)$$

Here  $C_{p_i}$  represents the concentration of Hb or BSA in the permeate solution at time  $t$ ; the concentration of Hb or BSA in the retentate solution is  $C_{f_i}$ . It is a function of the filtrate volume,  $V$ , collected over time  $t$ :

$$C_{f_i} = C_{f_0} - \frac{1}{V_0} \int_0^t C_{p_i}(t) dV(t) \quad (2)$$

where  $C_{f_0}$  is the starting concentration of Hb or BSA in the tank. The starting concentration of Hb is 0.2 mg/mL, and that of BSA is 1.0 mg/mL;  $V_0$  is the initial volume of feed solution in the tank, (200 mL); the total volume of solution filtered through the membrane up to time  $t$  is  $V(t)$ ;  $C_{p_i}(t)$  is the concentration of Hb or BSA in the permeate solution at any time  $t$ ; it is the same as  $C_{p_i}$ .

On dividing the volumetric filtration rate,  $Q$ , by the membrane area,  $A_m$ , the volume flux  $J_v$  is obtained (the unit of  $J_v$  used is  $\mu\text{m/s}$  ( $\approx 3.6$  LMH))

$$J_v = Q/A_m \quad (3)$$

To calculate the diavolume ( $N$ ) at any time  $t$ , equation (4) is used; it is the ratio of the total permeate volume at any time  $t$ ,  $V_{\text{totalp}}(t)$ , and the initial feed solution volume in the tank,  $V_{\text{tank}}$ :

$$N = \frac{V_{\text{totalp}}(t)}{V_{\text{tank}}} \quad (4)$$

The system performance is determined among others by the following quantities: purity of BSA in the tank (retentate); the recovery of Hb in the permeate; permeate purity of Hb; retention of BSA; selectivity of Hb over BSA. The Supplementary Information (SI) section provides the formulas for determining these quantities based on the concentrations of BSA and Hb in the retentate ( $C_{f\text{-BSA}}$ ,  $C_{f\text{-Hb}}$ ) and the permeate ( $C_{p\text{-BSA}}$ ,  $C_{p\text{-Hb}}$ ) at various times.

Selectivity,  $\Psi$ , of Hb and BSA at any time is calculated by the following equation:

$$\Psi = \left( \frac{C_{p\text{Hb}}}{C_{p\text{BSA}}} \right) / \left( \frac{C_{f\text{Hb}}}{C_{f\text{BSA}}} \right) = (S_{\text{Hb}} / S_{\text{BSA}}) \quad (5)$$

For the BSA-IgG separation system [20], essentially BSA in the above equations and those in the SI section is replaced by IgG and similarly Hb by BSA to obtain various quantities such as  $C_{f\text{-IgG}}$ ,  $C_{f\text{-BSA}}$ ,  $C_{p\text{-IgG}}$ , and  $C_{p\text{-BSA}}$  and corresponding rejections, purities, recoveries etc.

### 3. Results and discussion

#### 3.1. Hb-BSA separation studies - effect of cassette operational configurations

A major difference between a conventional cassette having one membrane on each side of the feed channel and an unchanged conventional cassette frame with three stacked membranes on each side for ISUF is a reduction in the feed channel height. This led to a significant increase in the feed flow pressure drop  $\Delta P_{\text{ff}}$  of as much as 68.95 kPa (10 psi) for a conventional feed flow rate of 30 mL/min. For Hb-BSA system, we found earlier [17] that optimum separation was obtained for a stack of 3 Ultracel® 100 kD membranes at a  $\Delta P_m$  (applied transmembrane pressure difference) of 10.35 kPa (1.5 psi), 2.3 mM Tris buffer and 7.8 pH. The experimental setup here (Fig. 1) allows permeate withdrawal from both ends. It would lead to a lower flow pressure drop  $\Delta P_{\text{fp}}$  in the

permeate side and correspondingly a higher  $\Delta P_m$  between the feed and the permeate at various locations.

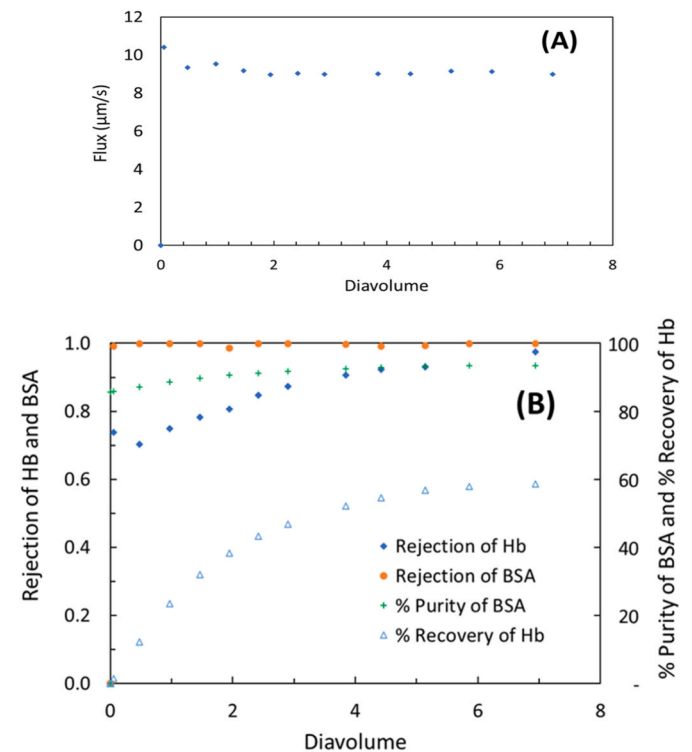
A way to avoid this increase in local  $\Delta P_m$  for the Hb-BSA separation system is to prevent permeate withdrawal from both ends and withdraw all of the permeate from the feed exiting end by shutting off the permeate-side valve next to pressure gage  $P_4$ . This increases the permeate flow pressure drop  $\Delta P_{\text{fp}}$  and therefore leads to a reduced value of local  $\Delta P_m$ s for any feed flow pressure drop profile. Results for such a configuration are shown in Fig. 2 for a retentate flow rate of 31.11 mL/min. Fig. 2A shows that the permeate volume flux for such a configuration is relatively stable around 9–10  $\mu\text{m/s}$  (32.4–36 LMH). The entering feed pressure is 55.16 kPag (8 psig) which explains the lower volume flux compared to that (11–13  $\mu\text{m/s}$  (39.6–46.8 LMH)) in a stirred cell at a lower  $\Delta P_m$  [17]. Exact comparison is difficult since here  $\Delta P_m$ s are varying along the length.

Fig. 2B illustrates the separation performance. First, it is clear that BSA rejection is almost complete over the whole diavolume range. Second, Hb rejection is high to start with and it increases with diavolume. The recovery of Hb increases with the diavolume but it plateaued out at around 60 %; correspondingly, purification of BSA also plateaued out at a similar diavolume.

The net effect of such a behavior on the purity of Hb in the permeate and the BSA retention is shown against diavolume in Fig. 3. The retention of BSA is very high around 99 % over 6 diavolumes and the purity of Hb in the permeate is maintained around 90 % over 2 diavolumes.

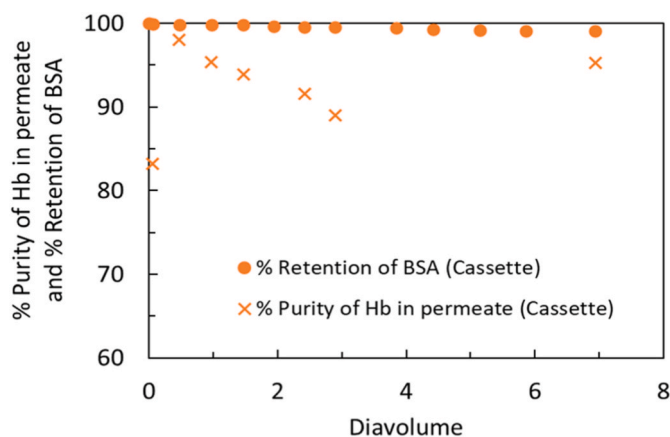
This is one of the highest performances in the cassette where purity of BSA reaches 93 % over 6 diavolumes (Fig. 2B), the rejection of BSA is almost complete, 99.99 % in the whole process, and the recovery of Hb reaches around 60 %.

A comparison of these results with those in a stirred cell [17] namely,



**Fig. 2.** Hb-BSA separation in a cassette as a function of diavolume when the valve next to permeate pressure gage  $P_4$  is closed: (A) Cassette flux; (B) Rejection behavior of Hb and BSA, purity of BSA in cell and recovery of Hb. Feed pressure  $P_1$ , 55.16 kPag (8 psig); retentate flow rate, 31.11 mL/min; permeate pressure  $P_3$  is 0 kPag (0 psig); permeate pressure  $P_4$  is 3.45 kPag (0.5 psig). Batch ultrafiltration feed: 1.0 mg/mL BSA and 0.2 mg/mL Hb, 2.3 mM Tris buffer: pH 7.8; 3 Ultracel® 100 kDa membranes in a stack.





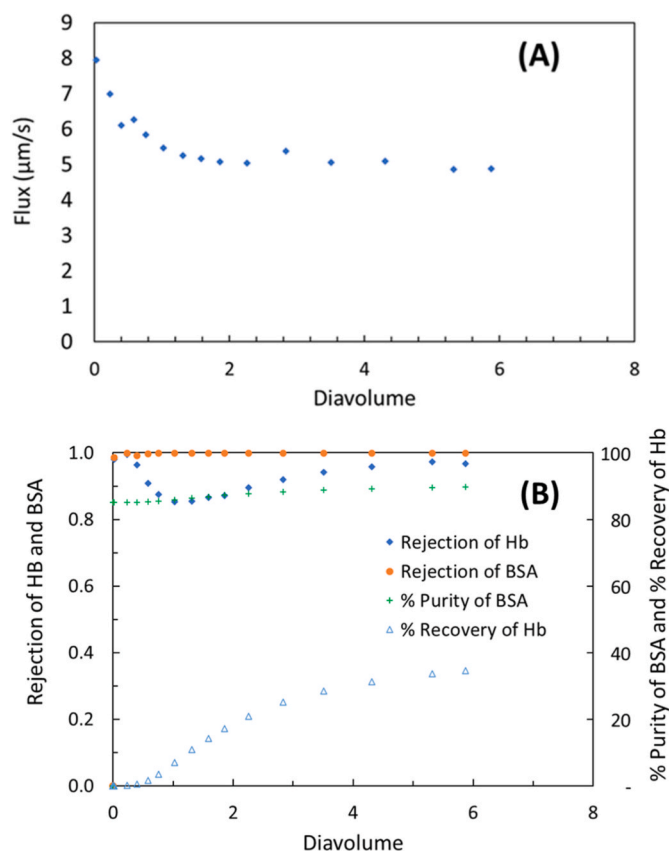
**Fig. 3.** Retention of BSA, purity of Hb in permeate as a function of diavolume when the valve next to permeate pressure gage  $P_4$  is closed. Feed pressure, 55.16 kPag (8 psig); retentate flow rate, 31.11 mL/min; the pressure gauge  $P_3$  at permeate is 0 kPag (0 psig); permeate pressure  $P_4$  is 3.45 kPag (0.5 psig). Batch ultrafiltration feed: 1.0 mg/mL BSA and 0.2 mg/mL Hb, 2.3 mM Tris buffer: pH 7.8; Ultracel® 100 kDa membrane.

near complete rejection of BSA ( $\sim 99.9\%$ ) and  $95\%$  + recovery of a highly purified Hb fraction in the permeate, is useful. The recovery of Hb is reduced here because of a higher  $\Delta P_m$  and not enough mixing leading to higher concentration polarization and a higher rejection of Hb. In the stirred cell studies [17], concentration polarization was much reduced also due to a much lower  $\Delta P_m$ . It is also necessary to recall the basis of such high separation in the stirred cell [17] where pH and membrane charge at low ionic strength played a critical role along with the 3-membrane stack. At pH 7.8, BSA carries a strong negative charge and is therefore strongly rejected by the negatively charged 100 kDa Ultracel® membrane with a further amplification by the 3-membrane stack. In contrast, Hb carried only a slight negative charge at this pH allowing it to permeate more easily through the membrane and faced reduced agglomeration with BSA.

An improvement over the behavior observed in Figs. 2 and 3 is shown in Fig. 4 where the system was operated at a lower feed flow rate of 20.72 mL/min leading to a lower entering feed pressure of 34.47 kPag (5 psig). Fig. 4A shows that the permeate volume flux is slowly getting reduced from 6 to about  $5 \mu\text{m/s}$  ( $\sim 18 \text{ LMH}$ ) over 6 diavolumes. This behavior is somewhat similar to that observed in a stirred cell [17]. Fig. 4B illustrates a very high rejection of BSA and a significantly higher rejection of Hb (compared to that in Fig. 2B) over a considerable range of diavolume. As a result, recovery of Hb is lower than that in Fig. 2B. Recovery of Hb keeps on increasing with diavolume to 40% and so does the purity of BSA with increasing diavolumes.

Consider now BSA retention, Hb purity and Hb-BSA selectivity for this data set via Fig. 5. In Fig. 5A, a very high BSA retention is observed over the whole DV range. Further, Hb purity as a result is also quite high reaching 98–99%. The calculated values of Hb-BSA selectivity for these performances are shown in Fig. 5B. After low values of selectivity for DV less than 0.5, the selectivity rises to as much as 1600 before dropping to  $\sim 350$  at a DV  $\sim 6$ . These values are high. Compared to all earlier results ( $\sim 70$ ) for this system [8] except that in a stirred cell ( $\sim 1000$ – $6000$  over 6 diavolumes) in the ISUF system [17] at a total  $\Delta P_m$  of 10.34 kPag. Selectivities calculated from Fig. 2B data (not shown) are significantly lower.

This performance enhancement (Fig. 5A and B) came about by reducing the feed pressure to 34.47 kPag and operating at a lower feed flow rate of 20.72 mL/min. These two variables are tied together to quite an extent since higher flow rate in the cassette leads to a higher feed pressure. Although a higher feed flow rate can reduce concentration polarization, higher pressure can increase polarization significantly. In a stirred cell, solute rejection in the Hb-BSA system was found to be



**Fig. 4.** Hb-BSA separation as a function of diavolume when the valve next to permeate pressure gage  $P_4$  is closed and feed pressure is lower: (A) Cassette flux; (B) Rejection behavior of Hb and BSA, purity of BSA in cell and recovery of Hb as a function of diavolume. Feed pressure,  $P_1$ , 34.47 kPag (5 psig); retentate flow rate, 20.72 mL/min; permeate pressure  $P_3$  is 0 kPag (0 psig). Batch ultrafiltration feed: 1.0 mg/mL BSA and 0.2 mg/mL Hb, 2.3 mM Tris buffer: pH 7.8; 3 Ultracel® 100 kDa membranes.

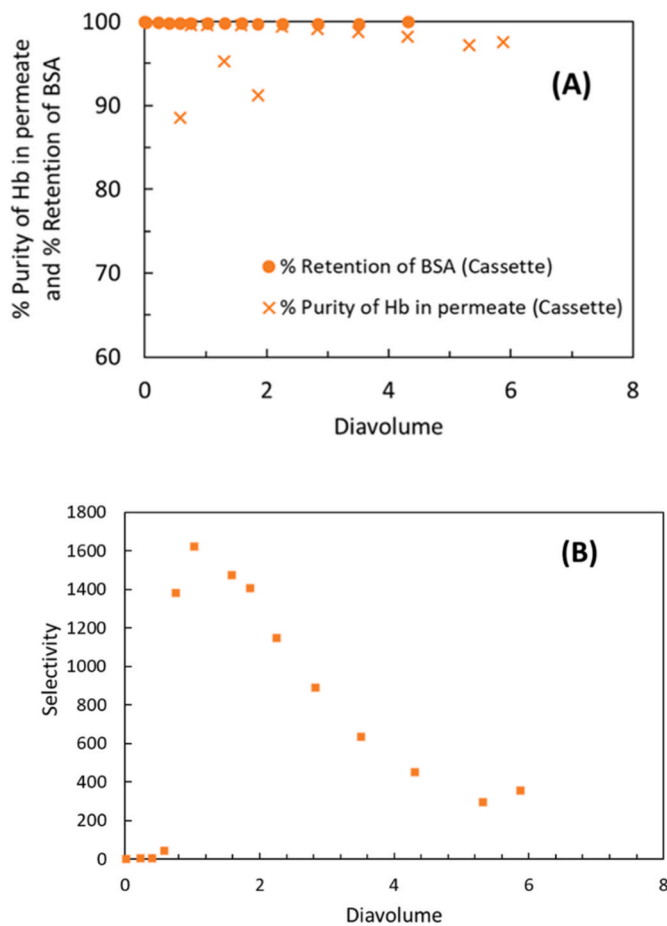
strongly affected by feed pressure; see Fig. 8 in Ref. [17].

We now focus on the role of permeate withdrawal configuration on the system performance. There are two outlets on the permeate side, one corresponding to the feed entrance location and the other located near the retentate exit location. The valve next to the pressure gage  $P_4$  located near the feed entrance location can be kept completely closed (as in the conditions for Figs. 2–5), completely open or half open. If the valve near the pressure gage  $P_4$  is left completely open in the permeate side, the behavior observed is shown in Fig. 6 for a retentate flow rate of 28.72 mL/min and a feed pressure of 55.16 kPag (8 psig).

Fig. 6A shows that the permeate volume flux goes through a considerable early drop-off of about  $2 \mu\text{m/s}$  ( $\sim 7.2 \text{ LMH}$ ) to a stable value  $\sim 9$ – $10 \mu\text{m/s}$  ( $\sim 32.4$ – $36 \text{ LMH}$ ). Fig. 6B shows that BSA rejection is not high enough; however, Hb rejection is quite high. This translates into a quite a low recovery of Hb which appears to reach a plateau of about 50%. Consequently, the purity of BSA increases at the beginning from 0.83 but becomes nearly constant at around 0.9. However, as shown in Fig. 7, retention of BSA drops to 93% over 6 diavolumes since the rejection of BSA is not high enough (around 97%); further, the purity of Hb in permeate is always lower than 60% over 6 diavolumes.

Fig. 8 illustrates the behavior of volume flux and the solute rejection behavior as a function of the diavolume used for the configuration where the valve next to the permeate side pressure gage  $P_4$  is half open. Fig. 8A shows first how the volume flux varies with increasing diavolume.

Fig. 8B focuses on various descriptions of the separation performance. The BSA rejection is decreasing from the very high values we have seen in Figs. 2–5. Correspondingly, purity of BSA hits a plateau

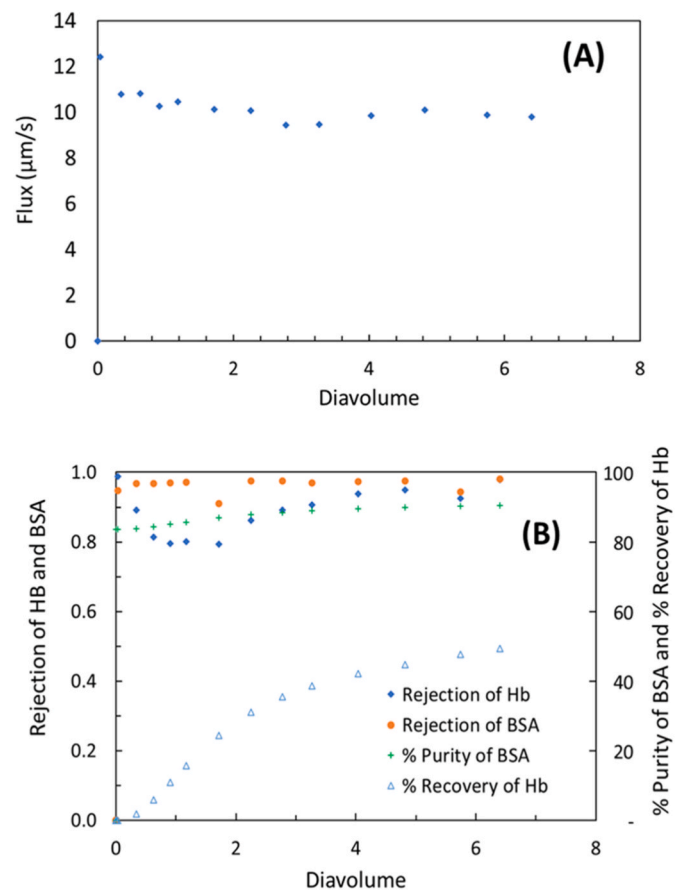


**Fig. 5.** BSA retention, Hb purity and Hb-BSA selectivity as a function of diavolume in 3-membrane configuration when the valve next to permeate pressure gage  $P_4$  is closed. A. BSA retention and Hb purity; B. Hb-BSA selectivity. Feed pressure,  $P_1$ , 34.47 kPag (5 psig); retentate flow rate, 20.72 mL/min; permeate pressure  $P_3$  is 0 kPag (0 psig). Batch ultrafiltration feed: 1.0 mg/mL BSA and 0.2 mg/mL Hb, 2.3 mM Tris buffer: pH 7.8; 3 Ultracel® 100 kDa membranes.

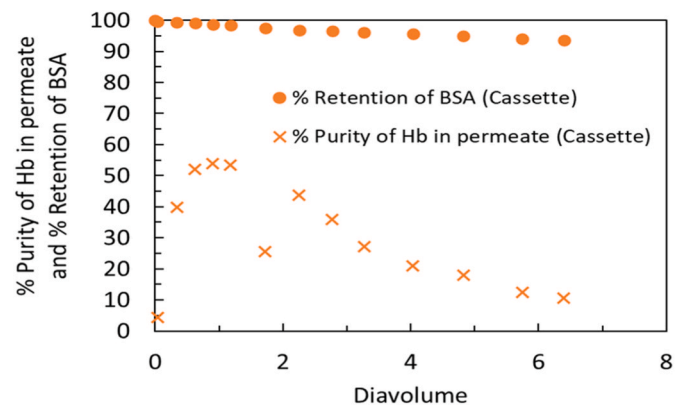
earlier. Rejection of Hb is high so that its purity will become lower in the permeate just as in Fig. 7.

On the basis of the observed separation and purification performances in Figs. 2–5, it is clear that very high purification and separation can be achieved for both BSA and Hb in the cassette having an ISUF 3-membrane stack of 100 kDa membranes. However, to obtain high recoveries, we need to operate over larger DVs. Slow plateauing of the Hb recovery in the cassette around 6 DV could signify difficulties not encountered in the stirred cell which operated under a significantly lower  $\Delta P_m$ . Therefore, a larger feed channel gap in the cassette will be desirable such that the feed flow pressure drop  $\Delta P_{ff}$  can be reduced considerably to the level encountered in commercially available cassette designs. This will facilitate maintaining a low value of  $\Delta P_m$  which is essential for the best performance in the Hb-BSA system.

The operating  $\Delta P_m$ s which allowed extraordinary separation and selectivity in Hb-BSA system in a stirred cell, are quite low [17]. However, systems studied earlier using the ISUF technique employed successfully much higher  $\Delta P_m$ s: (1) the system of myoglobin (17,566 Da) and  $\beta$ -lactoglobulin (35,500 Da in the form of a dimer) needed 68.94 kPag (10 psig) [10]; (2)  $\alpha$ -Lactalbumin (14,175 Da) and myoglobin (17,566 Da) system also needed 68.94 kPag (10 psig) [10]; (3) IgG-BSA system needed 31.25 kPag (4.5 psig) [20]. Cassette-based ISUF operation will therefore be doable for a variety of systems unburdened by the requirement of a low value of  $\Delta P_m$ .



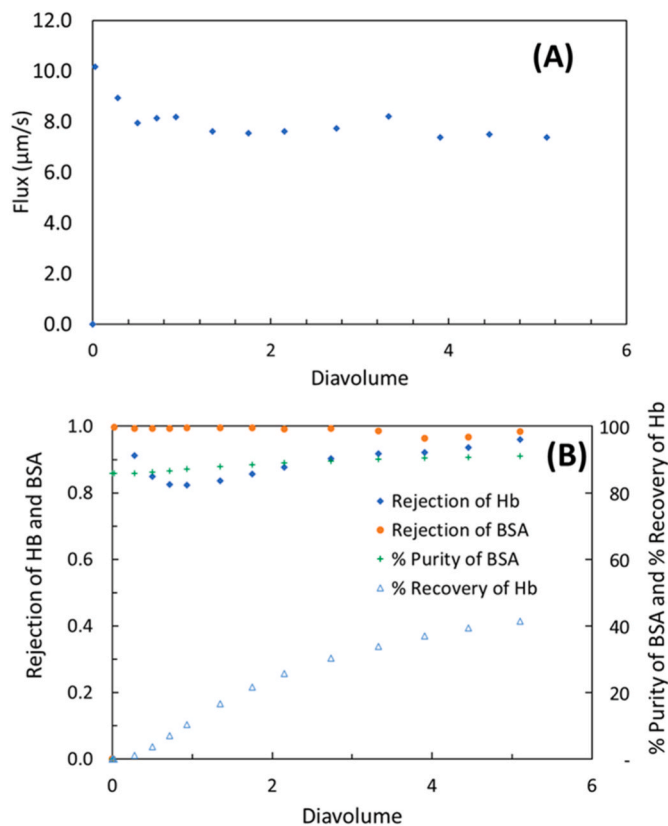
**Fig. 6.** Hb-BSA separation when valve near permeate pressure gage  $P_4$  is completely open: (A) Cassette flux; (B) Rejection behavior of Hb and BSA, purity of BSA in cell and recovery of Hb as a function of diavolume. Feed pressure, 55.16 kPag (8 psig); retentate flow rate, 28.63 mL/min; permeate pressure  $P_3$  is 0 kPag (0 psig). Batch ultrafiltration feed: 1.0 mg/mL BSA and 0.2 mg/mL Hb, 2.3 mM Tris buffer: pH 7.8; Ultracel® 100 kDa membrane.



**Fig. 7.** Retention of BSA, purity of Hb in permeate as a function of diavolume when the valve next to permeate pressure gage  $P_4$  is completely open. Feed pressure, 55.16 kPag (8 psig); retentate flow rate, 28.63 mL/min; the pressure gauge at permeate is 0 kPag (0 psig). Batch ultrafiltration feed: 1.0 mg/mL BSA and 0.2 mg/mL Hb, 2.3 mM Tris buffer: pH 7.8; Ultracel® 100 kDa membrane.

### 3.2. BSA-IgG separation studies - effect of cassette operational configurations

Following the Hb-BSA study, experiments were also conducted for a second system [20] namely, the separation of IgG and BSA where BSA is

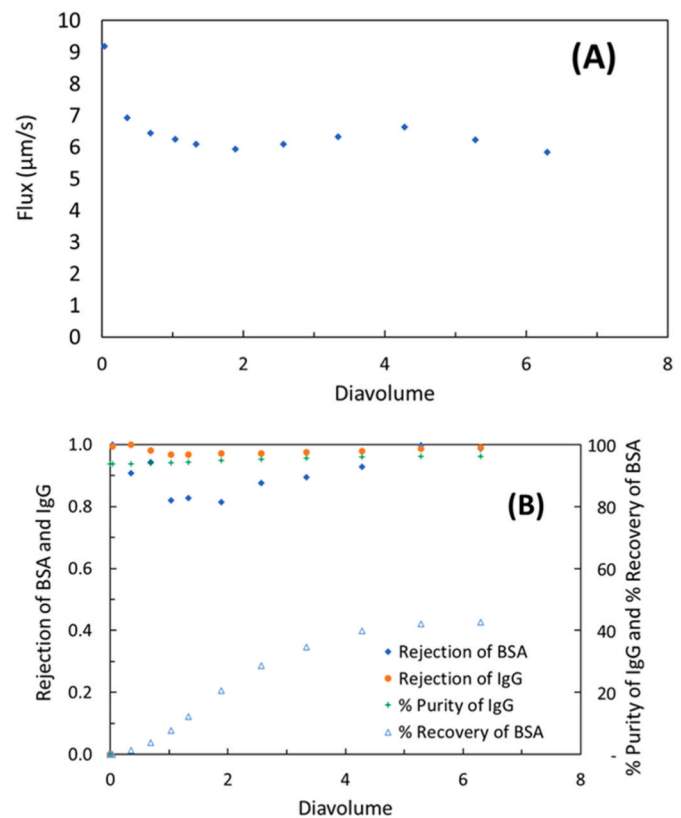


**Fig. 8.** Hb-BSA separation when the valve near permeate pressure gage  $P_4$  is half closed: (A) Cassette flux; (B) Rejection behavior of Hb and BSA, purity of BSA in cell and recovery of Hb as a function of diavolume. Feed pressure, 55.16 kPag (8 psig); retentate flow rate, 28.28 mL/min; permeate pressure  $P_3$  is 0 kPag (0 psig). Batch ultrafiltration feed: 1.0 mg/mL BSA and 0.2 mg/mL Hb, 2.3 mM Tris buffer: pH 7.8; Ultracel® 100 kDa membrane.

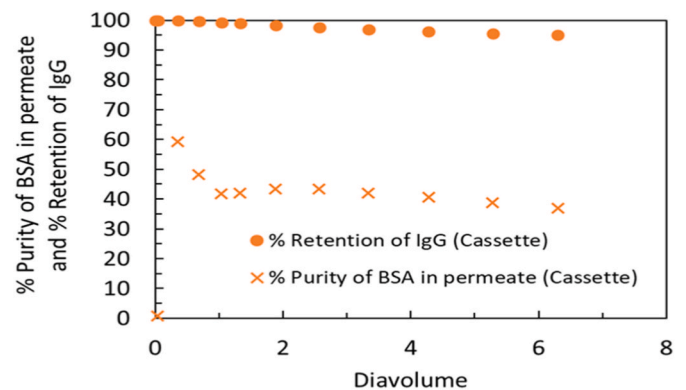
a model for HCPs in post-protein A eluate. All runs had: 1.0 mg/mL IgG, 0.1 mg/mL BSA, 2.3 mM acetate buffer, pH 4.8 (found optimal in Ref. [20]). This system's performances are presented in Figs. 9 and 10 for the condition where the valve near the pressure gage  $P_4$  was completely closed. As shown in Fig. 9, the permeate volume flux was approximately 5–6  $\mu\text{m/s}$  ( $\approx 18$ –21.6 LMH), which is lower than that observed during the BSA-IgG separation under the same operating conditions in a stirred cell [20]. The rejection of IgG was not high enough; it is around 95–98 % over the 1 to 2 DV range; however, BSA rejection was also relatively high, likely due to concentration polarization effects. Interestingly, both proteins exhibited similar rejection trends: a slight decline beyond 2 DVs and a rebound over 6 DVs.

Surprisingly, Fig. 10 reveals a significantly higher IgG retention compared to our earlier ISUF study with 3 Ultracel® membranes conducted in a stirred cell [20]. However, the purity of BSA in the permeate (around 40 % over 6 DVs) is much lower than that in the stirred cell studies where it went beyond 60 %. This suggests that the crossflow operation in the cassette system with the ISUF configuration enhanced BSA-IgG separation performance in particular ways relative to the dead-end mode in a stirred cell. In spite of that, a major challenge in the cassette-based ISUF system remains how to decrease BSA rejection and thereby increase BSA purity in the permeate.

An improvement over the performance of Fig. 9 was achieved when feed pressure was reduced to 34.47 kPag (5 psig) by operating at a lower feed flow rate of 20.85 mL/min as shown in Fig. 11 where we first show the volume flux in Fig. 11A. The volume flux is a bit lower than that in Fig. 9A reflecting a lower transmembrane pressure. Fig. 11B shows that IgG rejection continues to be lower (than almost perfect rejection) and similar to that in Fig. 9B. However, rejection of BSA is significantly

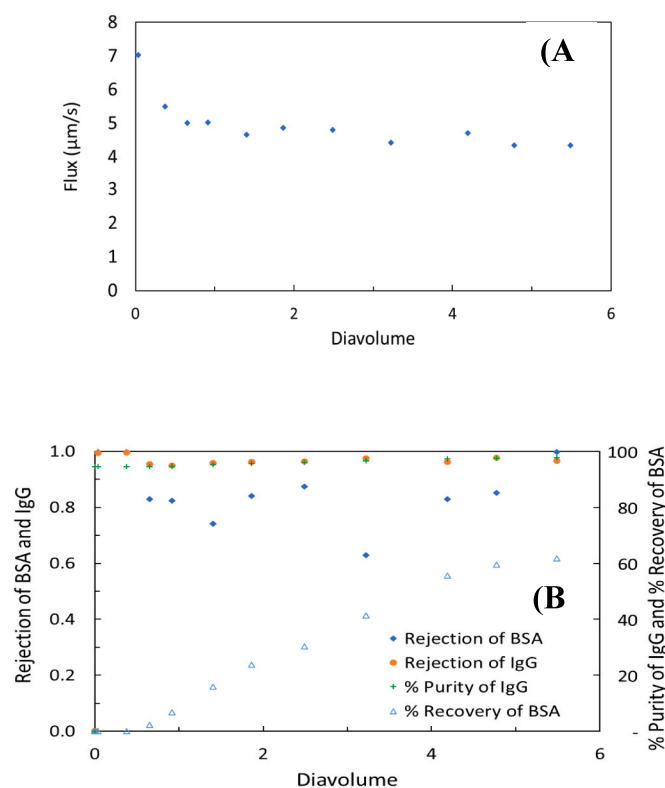


**Fig. 9.** BSA-IgG system: (A) Cassette flux in 3-membrane configuration; (B) Rejection of BSA and IgG, purity of IgG in cell and recovery of BSA as a function of diavolume in 3-membrane configuration. Feed pressure, 55.16 kPag (8 psig); flow rate of retentate, 29.51 mL/min. The valve in permeate side near permeate pressure gage  $P_4$  is closed. Batch ultrafiltration feed: 1.0 mg/mL IgG and 0.1 mg/mL BSA, 2.3 mM Acetate buffer: pH 4.8; Ultracel® 100 kDa membrane.

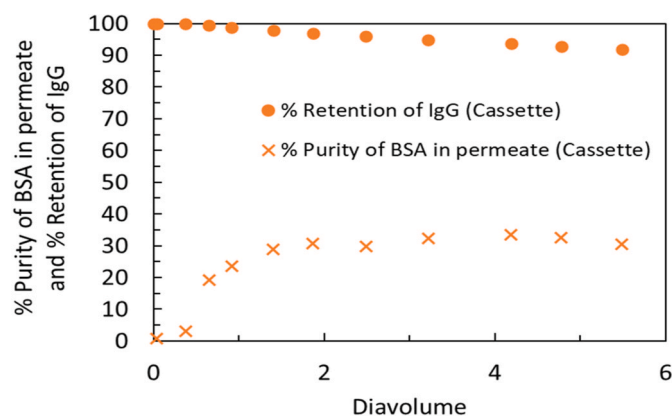


**Fig. 10.** BSA-IgG system: retention of IgG, purity of BSA in permeate as a function of diavolume in 3-membrane configuration for feed pressure, 55.16 kPag (8 psig); flow rate of retentate, 29.51 mL/min. The valve in permeate side near permeate pressure gage  $P_4$  is closed. Batch ultrafiltration feed: 1.0 mg/mL IgG and 0.1 mg/mL BSA, 2.3 mM Acetate buffer: pH 4.8; Ultracel® 100 kDa membrane.

lower than that in Fig. 9B. Correspondingly, BSA recovery is much higher just exceeding 60 %. Fig. 12 shows that IgG retention keeps going down with increasing diavolume as in Fig. 10. Further, BSA purity is low due to IgG leakage. We propose that for this system a lower transmembrane pressure difference and additional considerations mentioned below are important to achieve very high IgG retention via its almost complete rejection. The latter was achieved in Ref. [20]: 99.5 %



**Fig. 11.** BSA-IgG system at lower feed pressure: (A) Flux of cassette in 3-membrane configuration; (B) Rejection behavior of BSA and IgG, purity of IgG in cell and recovery of BSA as a function of diavolume in 3-membrane configuration. The pressure of entering feed is 34.47 kPag (5 psig), flow rate of retentate is 20.85 mL/min. The valve near  $P_4$  in permeate side is closed. Batch ultrafiltration feed: 1.0 mg/mL IgG and 0.1 mg/mL BSA, 2.3 mM Acetate buffer: pH 4.8; Ultracel® 100 kDa membrane.



**Fig. 12.** BSA-IgG system at lower feed pressure: Retention of IgG, purity of BSA in permeate as a function of diavolume in 3-membrane configuration. The pressure of entering feed is 34.47 kPag (5 psig), flow rate of retentate is 20.85 mL/min. The valve near  $P_4$  in permeate side is closed. Batch ultrafiltration feed: 1.0 mg/mL IgG and 0.1 mg/mL BSA, 2.3 mM Acetate buffer: pH 4.8; Ultracel® 100 kDa membrane.

retention and almost 100 % rejection of IgG.

The strategy adopted in stirred cell studies [20] for almost complete retention of IgG over 8–10 diavolumes along with high purification of IgG, high recovery of BSA with high purity involved using a modified ISUF technique where the 3-membrane stack involved two 100 kDa membranes supported by a 70 kDa membrane. This modified ISUF stack where different membranes were used broadened the capabilities of

ISUF for demanding separations such as no loss of IgG along with traditional expectations of high purity and high recovery of both species in two fractions. There is no particular reason why the cassette configuration cannot adopt a modified ISUF stack having a combination of different membranes.

#### 4. Concluding remarks

In recent literature, the internally-staged ultrafiltration (ISUF) technique of using a simple stack of three flat identical UF membranes in a stirred cell has been shown to effectively function almost as an UF membrane having a very narrow pore size distribution. This investigation has demonstrated that such an ISUF configuration can be scaled up to a cassette and function almost as effectively as in a stirred cell for mixtures that are difficult to separate by a single UF membrane. This was illustrated here for two binary protein mixtures: BSA-hemoglobin (MWR, 1.03); IgG-BSA (MWR, 2.26). However, in order to have as effective a performance as in a stirred cell over a wide range of diavolumes, the module frame design needs to accommodate the 3-membrane stack along with a channel height comparable to that in current cassettes. This can accommodate a broad range of transmembrane pressure drops by ensuring low values of feed flow pressure drop at feed flow rates needed to reduce concentration polarization. Demonstration of successful use of such a cassette for additional protein mixture separations will facilitate its acceptance in bioprocessing separation and purification applications.

#### CRediT authorship contribution statement

**Lixin Feng:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Yufeng Song:** Validation, Investigation, Formal analysis. **Gregory Straeffler:** Methodology, Conceptualization. **Andrew Bartlett:** Methodology, Conceptualization. **Christina Carbrello:** Writing – review & editing, Project administration. **Sagnik Basuray:** Writing – review & editing, Supervision, Funding acquisition. **Kamalesh K. Sirkar:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

#### Declaration of competing interest

The authors have no known competing financial interests or personal relationships that could influence the work reported here.

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#### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.memsci.2025.124658>.

#### Data availability

Data will be made available on request.



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